Strengths and Novelty of the Project

Investigating Genes Causing Male Infertility Using Bioinformatics Techniques

Unified report prepared for CVs and admission applications. Based on the project file and a literature review. Claims are phrased conservatively ("to our knowledge", "no identical prior study found"), because such wording withstands examination and peer review.

1. Short Version (about 150 words)

This is a large-scale retrospective in silico study that asks: are variants described as "pathogenic" in ClinVar actually linked to male infertility? From 37 candidate genes, 16 were retained with at least moderate gene–disease validity (Houston 2022; Zhao 2025). The project re-evaluated 712 pathogenic variants at the level of each ClinVar submission. Only 177 (24.9%) were recorded for male infertility; disease matching changed 6 of 15 representative variants, whereas threshold changes changed none. The same plan was executed along three tracks (Galaxy/VEP, Python on Colab, and six AI tools), with core values matching exactly (469/469 CADD scores and 131/131 allele frequencies). The tools were compared statistically against a documented reference. A bilingual website prototype presents evidence at the gene, variant, and recorded-disease levels.

Ready-to-use summary

"To our knowledge, this is among the first studies to re-evaluate small variants classified as pathogenic in public databases within validated male-infertility genes, requiring disease matching at the level of each submission, then executing the same protocol in three ways—graphical interface, programming, and AI tools—and comparing them statistically, while building a prototype website with a traceable, multi-layer evidence base."

2. Methodological Position, Idea, and Importance

- **Verification, not discovery.** The project does not provide new patient sequencing data and does not claim to discover new variants. It re-examines variants already submitted to public databases and interprets them within the male-infertility context. It is an interpretive layer that precedes molecular diagnosis and complements it; it does not replace it.
- **Reversing the usual question.** Instead of asking "is there a new variant?", the project asks: "are the variants we call pathogenic actually linked to infertility, or are they recorded for other diseases?" This inversion is the source of its epistemic value and distinguishes it from studies that merely rank variants by predicted harm.
- **The need.** Non-obstructive azoospermia (NOA) is one of the most severe forms of male infertility, and its genetic cause remains unknown in most cases. Houston et al. reported that only about 4% of infertile men receive a genetic diagnosis, while 60–70% remain unexplained. Lillepea et al. (2024) found a molecular diagnosis in about 12% (64/521). Zhou et al. (2025) estimated the pooled exome yield in NOA at about 15% (95% CI 10–20%; nine studies; 1,728 individuals; low certainty). Riera-Escamilla et al. (2025) suggested that using two classification approaches shows earlier studies may have overestimated diagnostic yield, highlighting the need for a unified, infertility-specific classification. This is the space the project addresses.
- **Honesty about limits.** The thesis explicitly states that it does not discover new variants, does not estimate diagnostic yield, and does not address copy-number changes or Y-chromosome deletions. This defines precisely what can and cannot be attributed to the project and prevents the overstatement that undermines otherwise good work in peer review.

3. The Decisive Separation of Three Concepts

The project separates three levels that are often conflated in variant tables and in AI-tool outputs:

- **Gene–disease validity:** is the gene itself causal? Oud et al. (2019) found only 78 genes confidently linked to male infertility out of 521 gene–disease relationships assessed.
- **Variant pathogenicity:** is this particular variant pathogenic, and for which disease was it recorded?
- **Variant priority for integrated assessment:** which variants deserve effort first?

Explicit inclusion criterion. A gene enters the database only with moderate or higher evidence under the ClinGen framework. A variant is considered a candidate only if it is pathogenic or likely pathogenic with at least one submitted assessment linking it to the infertility phenotype. Gene quality is therefore not confused with variant quality, and pathogenicity is not confused with disease association.

4. The Core Novelty: Disease Matching at the Submission Level

The project does not rely on ClinVar's aggregate classification of a variant. It goes down to each submitted assessment (SCV) and asks: for which disease did this submitter record the variant—male infertility or another disease? This is the most important novelty of the project. It is necessary because a single variant may be recorded for spermatogenic failure and premature ovarian failure, as documented in ClinVar.

Numbers reported in the project

- From 6,157 ClinVar variants in the candidate genes → 726 pathogenic or likely pathogenic → 712 in the 16 validated genes.
- Of the 712, only 177 (24.9%) were recorded for male infertility; about three quarters were not recorded for male infertility.
- In another analysis with a denominator of 734 variants: 175 (23.8%) had any evidence of an infertility association, and 477 (65.0%) were recorded for other diseases only.
- In the first five-gene route: 38 variants matched infertility (6.6%), 11 were mixed, 466 were non-matched (80.5%), and 64 were unspecified.

Examples showing that gene strength does not transfer to all variants

- **AR (the gene with the highest evidence for infertility):** only 7 of 385 variants (1.8%) were recorded for infertility in one project table; in another table, 340 of 397 pathogenic variants (85.6%) were recorded for other diseases only, chiefly androgen insensitivity syndrome and Kennedy disease.
- **XRCC2:** all 17 variants (17/17) were recorded for other diseases, 15 of them linked to cancer predisposition; it had no disease-matched representative.
- **In contrast,** all variants of TEX11, TEX14, SYCP2, M1AP, and TERB1 were recorded for infertility. The rule does not generalize one judgment to all genes; it shows which genes actually contribute to the infertility phenotype.

This is consistent with the literature on public databases. Shah et al. (2018) found apparent inflation in pathogenic variant sets relative to disease prevalence, reaching 11.5% of disorders in pathogenic sets and 92.3% in the conflicting-interpretation set. Other studies addressed the causes of conflicting interpretations in ClinVar (Lazareva et al. 2024) and the high proportion of downgrades when common variants are reinterpreted (Xiang et al. 2020).

"Highest CADD alone" is not enough

Many analyses rank variants by CADD and choose the highest. The project tested this directly: this rule alone would select 6 of 15 representatives (40%) recorded for other diseases, whereas moving the CADD threshold from 15 to 25 changed none of them. In the missense category, 4 of 14 highest-CADD selections were not recorded for male infertility; in NR5A1, a variant with CADD 35 was excluded because its recorded disease was a mixed syndrome. The score measures general probability of harm, not relevance to a specific disease, and this appeared in the project's own data.

5. Methodological Stability

- **Groups defined before results:** group H (mechanical rule: highest CADD without a disease filter), group C (highest CADD among disease-matched variants), and group M (strongest missense variant per gene). This resembles pre-registration and prevents selecting results after seeing them.
- **Nine threshold scenarios** (CADD 15, 20, 25; gnomAD 0.0001, 0.001, 0.01): all 15 representatives remained stable. What changes with the threshold is the size of the candidate list, not the identity of the representatives.
- **The rarity threshold does not create the list:** 549 of 727 variants were absent from gnomAD, and the threshold excluded only one variant (MEI1 at 0.002171).
- **The first two tracks matched perfectly** in core values (469/469 CADD scores and 131/131 allele frequencies), and the unified analysis reproduced all 15 representatives.
- Pre-definition and sensitivity testing are rare in graduation projects; they separate an analysis that "looks good" from one that is reproducible and testable.

6. Technical and Engineering Strength

- **Cell-to-source traceability:** an Excel matrix with INDEX/MATCH formulas that pull values from source sheets (48 sheets), so no value is manually copied and untraceable.
- **Recomputing expression from raw files** (GTEx v8 and HPA) revealed documented errors: MSH5 in testis was 3.10 and was corrected to 34.60; AR was 32.40 and was corrected to 7.02.
- **Reproducibility structure:** SHA-256 fingerprints, a rerunnable Colab notebook, read-only inputs, no overwriting of previous-stage outputs, every correction in a new folder, and no substitute value on failure.
- **Reference and structural verification:** MANE Select was used, and the amino acid was checked in structural models. A numbering shift in TEX11 and SHOC1 was detected, and SHOC1 was not interpreted rather than giving a wrong interpretation.
- **Integration of 13 data layers in one matrix:** ClinVar (SCV), CADD v1.7, gnomAD v4.1.1, SpliceAI, REVEL, AlphaMissense, LOFTEE, NMD, 1000 Genomes, OMIM, GTEx v8, HPA (RNA and immunostaining), and AlphaFold/UniProt.
- **Missing values handled in three states:** "not observed", "technically unavailable", and "not applicable", without imputation or coordinate inference.

7. Missense Variants, Structural Analysis, and NMD

- Because the selection rule favours high-scoring stop variants, an expansion was dedicated to selecting the strongest missense variant per gene in two forms: highest CADD alone, and CADD with disease matching.

- About 13 missense variants were displayed in ChimeraX and PyMOL over AlphaFold models with pLDDT values, together with reproducible command files. This is a descriptive, hypothesis-generating analysis; it does not prove pathogenicity.
- NMD was predicted for all eight stop variants; this is a computational prediction, not functional proof.
- The final set contains 15 representatives in 16 genes (8 stop, 5 splice, 2 missense), with CADD scores from 27.2 to 46; six are absent from gnomAD and the remaining nine have frequencies no higher than 0.00007.
- The most frequent genes in Zhou (2025), such as AR, TEX11, M1AP, and DMRT1, overlap with the project set; NR5A1 was the most frequent in Lillepea (2024). This is external consistency, not a product of the rule alone.

8. Statistical Strength

- Tests were chosen to match paired data: Cochran's Q for overall comparison, exact McNemar for pairwise comparisons, Holm correction for multiple comparisons, Wilson intervals for proportions, the Jaccard index for overlap, and a gene-clustered bootstrap with 3,000 resamples (seed 20261012).
- Cochran's Q for four tools: H: $Q = 9.790$, $p = 0.0204$; C: $Q = 7.828$, $p = 0.0497$; M: $Q = 14.087$, $p = 0.0028$. For three tools in M: $p = 0.0023$.
- Of 18 pairwise comparisons, four had a raw $p < 0.05$, and none survived Holm correction (smallest corrected $p = 0.1143$).
- Non-significance after correction was reported rather than hidden. The likely explanation is the small sample size (20–32 variants per group) combined with multiple comparisons.
- Tools were not ranked by "clinical accuracy" because no independent gold standard exists. They were ranked by one declared criterion: agreement with the documented internal reference.

9. Working with AI: A Methodological Contribution

- Six tools/models received the same plan: Claude (Opus 5.5), Devin, and Gemini 3 Pro in a single command; ChatGPT Plus and OpenCode stage by stage; GPT-6 Astra, ChatGPT Plus, and OpenCode were assigned a 14-variant verification task. They were compared with two conventional references: Galaxy and Google Colab.
- **Independence audit with digital fingerprints:** SHA-256 showed that 21 of 22 files in the first mission between OpenCode and ChatGPT Plus were byte-identical, meaning they were not independent runs; any agreement between them is not independent agreement.
- **Reading raw files reveals memory errors:** Gemini contradicted the raw GTEx file in 14 of 14 genes (16 of 16 contradictory claims; Wilson 80.6% to 100%) and attributed a non-existent classification to MFGA. OpenCode recomputed GTEx from the raw file and exposed the discrepancies.
- **Documenting tool limits:** GPT-6 Astra could not retrieve GTEx, OMIM, and VarCards2 (0 of 14; Wilson 0 to 21.5%). This was documented as an execution-environment limitation, not a biological result—a distinction between "I did not find" and "it does not exist".
- **Version instability:** Claude's outputs changed between versions (14 vs 18 vs 15 in set M), so the version must be fixed with every result.
- **A proposed six-step protocol** for comparing AI tools in genomics: a unified candidate list, an explicit decision per tool, versions and fingerprints, independent environments, an independent reference standard, and pre-declared weights.

- The conclusion is that AI tools are useful for cross-checking if their outputs are verified against primary sources; they are not a source of truth.

10. Website and Evidence Base

The prototype. A bilingual (Arabic/English) static website that works without a server or internet connection, with six sections: chromosome map, gene cards, a "selection lab" that reapplies the rule when the user changes thresholds, a statistical analysis workspace, a structural image gallery, and database links. It includes a JavaScript engine that recomputes Cochran's Q, McNemar, Holm, Bonferroni, Wilson, Jaccard, and bootstrap in the browser; its values matched Python. This turns the database from an archive into a "selection lab" whose claims the reader can test.

The multi-layer, multi-track evidence base. The database is not a "gene list" but a variant-level evidence matrix with independently sourced layers:

- **Genomic layer:** ClinVar at SCV level, CADD, gnomAD, and 1000 Genomes: is the variant rare? How harmful is it? For which disease was it recorded?
- **Tissue-expression layer:** GTEx and HPA: is the gene expressed in the testis?
- **Functional layer:** SpliceAI, REVEL, AlphaMissense, LOFTEE, and NMD: does the variant affect splicing or the protein?
- **Structural layer:** AlphaFold, pLDDT, ChimeraX, PyMOL, and OMIM: where is the substitution, and does it fit the phenotype and mode of inheritance?

Why it is not an arbitrary database. A gene enters only with moderate or higher ClinGen evidence; a variant enters only if it is pathogenic or likely pathogenic with a submission linking it to the infertility phenotype; layers come from raw files, not summaries; and the database can be updated with each ClinVar release. It is described as "evidence-based, reviewable, and updatable", not "definitively proven"; no database in the world is 100% proven to cause infertility, and the project itself demonstrated this with its numbers.

Comparison with existing databases. Published male-infertility databases include the Male Fertility Gene Atlas (MFGA), the Male Infertility Knowledgebase, IDDB, FertilityOnline, and SpermatogenesisOnline; some integrate tissue and testicular-cell expression and variant statistics. Adding expression layers is therefore not the new element. The unoccupied space, to our knowledge, is a variant-level interpretive database specialized in male infertility, matching disease at the SCV level, tracing every value to its source and to more than one execution track, and allowing the user to reapply the rule with custom thresholds. The current prototype is limited to this study's results (16 genes) and is intended for later expansion.

11. Verification of "First" Claims

The phrase "no tool has been evaluated on a complete classification track against a verified reference" is too broad, because ClawBench and others evaluated this for ACMG classification in general. The correct wording is: we found no evaluation of AI tools on a complete prioritization plan for a specific rare disease against a reference documented from primary sources, with paired inferential tests.

12. Clinical Path and Practical Output

- In the molecular-diagnosis gap, the project provides an interpretive layer that re-examines submitted variants and links them to the infertility phenotype; it complements clinical sequencing, not replaces it.

- If about three quarters of pathogenic variants in these genes are not recorded for infertility, re-matching may prevent attributing a cancer variant or an androgen syndrome to infertility and redirect the case to the right specialty.
- The output is a disease-matched candidate list ready for functional validation, not a diagnosis. The distinction between "pathogenic in ClinVar" and "cause of infertility", and between "candidate" and "diagnosis", is preserved.

13. Strength as a University Graduation Project

- **Scale and scope:** six chapters, 34 tables, 44 figures, more than 130 references, a 48-sheet Excel matrix, and a log of 502 audited files.
- **Multiple skills:** bioinformatics, Python programming, inferential statistics, structural analysis, web development, and AI-tool evaluation in one coherent project.
- **Scientific maturity:** the thesis criticizes itself, documents a substantive correction (changing the NR5A1 representative), discovers and corrects expression-value errors, and presents limitations in a dedicated table.
- **Publishability:** at least two papers can be extracted: a methodological paper on disease matching and ClinVar reinterpretation in male infertility, and a paper on evaluating AI tools in genomics with the comparison protocol.

14. Limitations (Which Strengthen Credibility)

- A retrospective computational study on public data; it collects no patient samples, discovers no new variants, and does not estimate diagnostic yield.
- It is limited to small single-gene variants (SNV and indel); it does not cover chromosomal abnormalities, Y-chromosome deletions, or copy-number changes.
- The thresholds (CADD ≥ 20 and gnomAD frequency ≤ 0.001) are conventional in rare-disease studies, not mathematically derived for this disease, and their sensitivity was tested.
- Structural analysis and NMD prediction generate hypotheses; they do not prove causality.
- AI outputs are not an independent source, and the comparison results apply to the studied groups only; they are not generalized to the tools beyond those groups.
- No database is 100% proven, so the resource is described as evidence-based and revisable.

15. Conclusion and Strongest Sentence

The project has a defensible novelty in a combination of elements: SCV-level disease matching, multiple evidence layers from raw files, a paired statistical comparison between bioinformatics and AI tracks with SHA-256 independence auditing, and an interactive website that recomputes the tests. These elements have not been combined in this way in male infertility, to our knowledge.

The strongest result in one sentence: about three quarters of variants classified as pathogenic in validated male-infertility genes are not recorded for male infertility; "highest CADD" alone would select 6 of 15 representatives recorded for other diseases; no AI difference survived Holm correction; and the tools that read raw files were the ones that caught the errors.

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